Lecture 34

Second-order linear equations

Uses.

Summary 8: Product and chain rules and the elementary derivative table hold on their stated domains.
Summary 9: A real derivative which is identically zero on an interval gives a constant function.
Summary 16: Complex arithmetic, the exponential product law, and $(e^{\lambda t})' = \lambda e^{\lambda t}$ apply with real time and complex constants.
Summary 21: A 2×2 linear system has a unique solution when its determinant is nonzero.
Summary 29: The ratio and comparison tests prove absolute convergence, and $nq^n \to 0$ for $|q| < 1$.
Summary 31: Inside their radius, real power series may be differentiated termwise. Complex convergence is equivalent to component convergence.
Summary 33: The unique complex-valued solution of $z' = \lambda z$, $z(0) = z_0$, is $z_0 e^{\lambda t}$; the proof is independent of the cited existence theorem.

Objectives. *Characteristic roots provide solutions of a constant-coefficient equation. Factorisation proves that these solutions give every solution, including the repeated-root case. Initial position and velocity determine their coefficients. The same argument treats damping and exponential forcing, and first-order uniqueness identifies the exponential series for every complex argument.*

By the end of this lecture you will be able to:

- Prove completeness of the characteristic-root formulas for a homogeneous scalar second-order equation with constant coefficients.
- Determine the unique real solution from two real initial data in each discriminant case.
- Derive damping and exponential-forcing formulas, including repeated characteristic roots.
- Prove the factorial-series representation of the complex exponential using real-parameter uniqueness.

34.1 The characteristic equation

Let $p, q \in \mathbb{R}$ be constants. A complex-valued function y on an open real interval I solves the homogeneous second-order equation when it is C^2 componentwise and satisfies

$$y'' + py' + qy = 0 \qquad (t \in I). \tag{34.1}$$

Real-valued solutions are included. An initial-value problem prescribes $y(0) = y_0$ and $y'(0) = v_0$, with $0 \in I$. The *characteristic polynomial* is $P(\lambda) = \lambda^2 + p\lambda + q$. Substitution of $e^{\lambda t}$ gives

$$(e^{\lambda t})'' + p(e^{\lambda t})' + qe^{\lambda t} = P(\lambda)e^{\lambda t}.$$

The exponential never vanishes, so this function is a solution exactly when $P(\lambda) = 0$. The quadratic formula over $\mathbb{C}$ gives two roots counted with multiplicity, written λ_1 and λ_2. Their sum is $-p$ and their product is q.

Finding two solutions by substitution does not establish completeness. We prove that every solution has the proposed form by reducing the equation to two first-order equations.

Theorem 34.1 (Complete homogeneous solution). *On any open interval I containing zero, every complex-valued solution of Eq. (34.1) has exactly one of the following forms, with unique constants $c_1, c_2 \in \mathbb{C}$:*

$$\begin{array}{c|l} \lambda_1 \neq \lambda_2 & y(t) = c_1 e^{\lambda_1 t} + c_2 e^{\lambda_2 t} \\ \lambda_1 = \lambda_2 = \lambda & y(t) = (c_1 + c_2 t)e^{\lambda t}. \end{array}$$

Every function in the indicated family is a solution. Any prescribed pair $y_0, v_0 \in \mathbb{C}$ determines exactly one solution on $\mathbb{R}$.

Proof. Let y be a solution and set $w = y' - \lambda_2 y$. The characteristic sum and product identities give

$$w' - \lambda_1 w = y'' - (\lambda_1 + \lambda_2)y' + \lambda_1\lambda_2 y = y'' + py' + qy = 0.$$

By Lecture 33's complex first-order uniqueness, $w(t) = Ce^{\lambda_1 t}$ for the constant $C = w(0)$. The product rule consequently gives

$$(e^{-\lambda_2 t}y)' = Ce^{(\lambda_1 - \lambda_2)t}.$$

If the roots are distinct, a primitive of the right side is $Ce^{(\lambda_1-\lambda_2)t}/(\lambda_1 - \lambda_2)$. Subtracting this primitive leaves a complex function with zero derivative. Its real and imaginary parts are constant by Lecture 9. Thus $y = c_1 e^{\lambda_1 t} + c_2 e^{\lambda_2 t}$, with $c_1 = C/(\lambda_1 - \lambda_2)$ and some $c_2 \in \mathbb{C}$. If the roots coincide, the right side is C and the primitive is Ct. The same constant-difference argument gives $y = (c_1 + c_2 t)e^{\lambda t}$. This proves completeness on I.

Conversely, each exponential for a root solves the equation by the substitution above, and finite linearity preserves the equation. At a double root, direct differentiation gives $(D - \lambda)((c_1 + c_2 t)e^{\lambda t}) = c_2 e^{\lambda t}$, where D denotes differentiation with respect to t. Applying $D - \lambda$ once more gives zero. Since $(D - \lambda)^2 = D^2 + pD + q$ for a double root, these functions also solve the equation.

For distinct roots the initial data require the system

$$\begin{pmatrix} 1 & 1 \\ \lambda_1 & \lambda_2 \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = \begin{pmatrix} y_0 \\ v_0 \end{pmatrix}.$$

Its determinant is $\lambda_2 - \lambda_1 \neq 0$. The unique coefficients can also be obtained by subtraction:

$$c_1 = \frac{v_0 - \lambda_2 y_0}{\lambda_1 - \lambda_2}, \qquad c_2 = \frac{\lambda_1 y_0 - v_0}{\lambda_1 - \lambda_2}.$$

At a double root the data require

$$\begin{pmatrix} 1 & 0 \\ \lambda & 1 \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = \begin{pmatrix} y_0 \\ v_0 \end{pmatrix}, \qquad c_1 = y_0, \quad c_2 = v_0 - \lambda y_0.$$

This determinant is 1. Both families are defined on $\mathbb{R}$, so these coefficients supply global solutions and uniqueness on every interval through zero. ∎

The notation $(D - \lambda_1)(D - \lambda_2)$ is a factorisation of a differential expression: applying the two first-order expressions gives the stated second-order expression. The factors commute here because the coefficients are constant. No claim about varying coefficients is needed.

Example 34.2 (Two distinct real roots)**.** Solve $y'' - 3y' + 2y = 0$, $y(0) = 2$, $y'(0) = 1$. The polynomial is $P(\lambda) = (\lambda - 1)(\lambda - 2)$. Completeness gives $y = c_1 e^t + c_2 e^{2t}$. The initial equations and their solution are

$$c_1 + c_2 = 2, \qquad c_1 + 2c_2 = 1, \qquad c_1 = 3, \quad c_2 = -1.$$

Thus $y(t) = 3e^t - e^{2t}$ is the unique solution on $\mathbb{R}$. The value changes sign at $t = \ln 3$, since $y = e^t(3 - e^t)$ and $e^t > 0$. Both roots are required to attain the specified position and derivative.

34.2 Real solutions in the three root cases

Suppose now that $y_0, v_0 \in \mathbb{R}$ and define the discriminant $\Delta = p^2 - 4q$. If $\Delta > 0$, the roots are distinct and real, so the coefficient formulas show $c_1, c_2 \in \mathbb{R}$. If $\Delta = 0$, the repeated root $\lambda = -p/2$ is real and the coefficients y_0 and $v_0 - \lambda y_0$ are real.

If $\Delta < 0$, write the roots as $\alpha + i\beta$ and $\alpha - i\beta$, where $\alpha = -p/2$ and $\beta = \sqrt{-\Delta}/2 > 0$. The exponential identities of Lecture 16 suggest the real expression

$$y(t) = e^{\alpha t}(A \cos \beta t + B \sin \beta t).$$

The functions $e^{\alpha t} \cos \beta t$ and $e^{\alpha t} \sin \beta t$ are linear combinations of the two characteristic exponentials, so both functions solve the equation. Its initial data give

$$A = y_0, \qquad \alpha A + \beta B = v_0, \qquad B = \frac{v_0 - \alpha y_0}{\beta}.$$

This constructs a real solution with the required data. The uniqueness already proved over $\mathbb{C}$ makes every other solution with those data equal to it, so no additional real solutions remain. The real families are therefore

$\Delta > 0$	$c_1 e^{\lambda_1 t} + c_2 e^{\lambda_2 t} \quad (c_1, c_2 \in \mathbb{R})$
$\Delta = 0$	$(c_1 + c_2 t) e^{-pt/2} \quad (c_1, c_2 \in \mathbb{R})$
$\Delta < 0$	$e^{-pt/2}(A \cos \beta t + B \sin \beta t) \quad (A, B \in \mathbb{R}).$

For example, $y'' + \omega^2 y = 0$ with $\omega > 0$ has the complete family $A \cos \omega t + B \sin \omega t$. Initial values y_0, v_0 give $A = y_0$ and $B = v_0/\omega$. The division requires $\omega > 0$; for $y'' = 0$ the repeated-root formula instead gives $y = y_0 + v_0 t$.

34.3 Damping of a spring motion

Assume the prescribed model for displacement x is

$$mx'' + cx' + kx = 0, \qquad m, k > 0, \quad c \geq 0.$$

The term $-kx$ is the restoring force and $-cx'$ opposes velocity. Define $\gamma = c/(2m)$ and the positive undamped angular frequency $\omega_0 = \sqrt{k/m}$. The normalised characteristic polynomial has roots $-\gamma \pm \sqrt{\gamma^2 - \omega_0^2}$. The cases are distinguished by comparing γ and ω_0:

$$\begin{array}{c|c} 0 \leq \gamma < \omega_0 & x = e^{-\gamma t}(A\cos\omega_d t + B\sin\omega_d t), \quad \omega_d = \sqrt{\omega_0^2 - \gamma^2} > 0 \\ \gamma = \omega_0 & x = (A + Bt)e^{-\gamma t} \\ \gamma > \omega_0 & x = Ae^{(-\gamma+\sqrt{\gamma^2-\omega_0^2})t} + Be^{(-\gamma-\sqrt{\gamma^2-\omega_0^2})t}. \end{array}$$

These are called *underdamped*, *critically damped* and *overdamped*, respectively. The first case includes the undamped case $c = 0$. In the overdamped case both roots are negative, since $0 < \sqrt{\gamma^2 - \omega_0^2} < \gamma$. When $c > 0$, every displacement in these families tends to zero as $t \to \infty$. The critical term also decays: for $n \leq t < n+1$, $0 \leq te^{-\gamma t} \leq (n+1)(e^{-\gamma})^n \to 0$ by Lecture 29. At $c = 0$, nonzero oscillations persist.

Figure 34.1 compares three motions from the same initial data, with $m = k = 1$ and damping coefficients $c = 1$, 2 and 3.

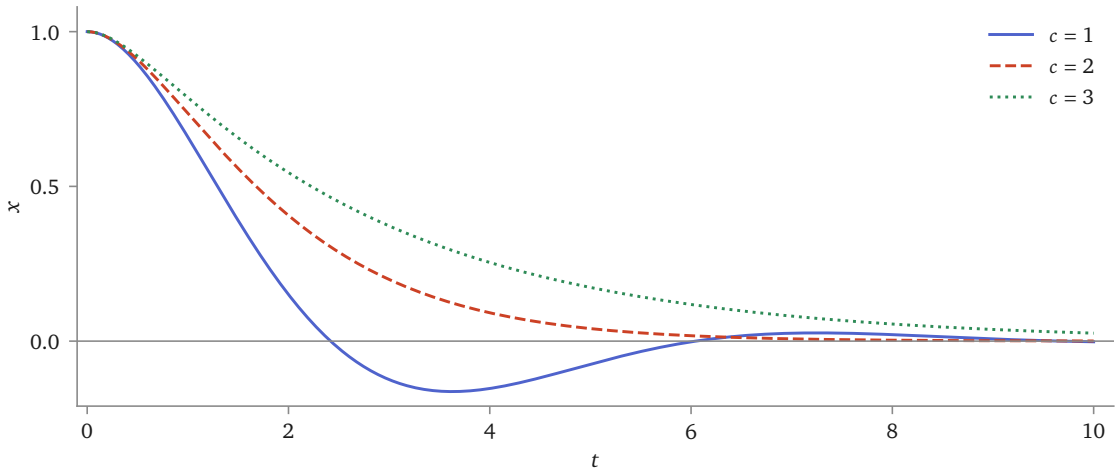


Figure 34.1: Exact displacements with $x(0) = 1$ and $x'(0) = 0$. The three curves have $c = 1$ (underdamped), $c = 2$ (critical) and $c = 3$ (overdamped), with the same mass and spring constant.

Example 34.3 (The extra solution at critical damping)**.** Take $m = 1$, $c = 4$, $k = 4$ and initial values $x(0) = 1$, $x'(0) = 0$. The characteristic polynomial is $(\lambda + 2)^2$. The repeated-root family and initial data give

$$x = (A + Bt)e^{-2t}, \qquad A = 1, \qquad B - 2A = 0, \qquad x(t) = (1 + 2t)e^{-2t}.$$

Using only Ae^{-2t} would force $x'(0) = -2x(0)$ and could not attain the given data. The term te^{-2t} supplies the missing independent initial derivative. The actual derivative is $x' = -4te^{-2t}$. Thus for $t \geq 0$ the displacement stays positive and decreases to zero, with its horizontal tangent at the initial time.

The mechanical energy of this model is defined by $E(t) = \frac{1}{2}m(x')^2 + \frac{1}{2}kx^2$. Both terms are nonnegative and differentiation gives

$$E' = x'(mx'' + kx) = -c(x')^2 \le 0.$$

Thus the model loses energy where velocity is nonzero and $c > 0$. For $c = 0$ the energy is constant. The explicit root formulas, rather than energy monotonicity alone, established decay of the displacement above.

34.4 Exponential forcing and characteristic roots

Consider $y'' + py' + qy = Fe^{\sigma t}$, where $F, \sigma \in \mathbb{C}$ are constants. A *particular solution* is any one function satisfying this inhomogeneous equation. Subtracting a particular solution from any other solution gives a homogeneous solution by linearity. Consequently the general solution is the particular solution plus the complete homogeneous family. Its coefficients are fixed by subtracting the particular solution's two initial values from the prescribed data.

To construct a particular solution, write $y = e^{\sigma t}u(t)$. The product rule twice gives the identity

$$y'' + py' + qy = e^{\sigma t}\big(u'' + (2\sigma + p)u' + P(\sigma)u\big). \tag{34.2}$$

For example, the first derivative is $e^{\sigma t}(u' + \sigma u)$ and the second is $e^{\sigma t}(u'' + 2\sigma u' + \sigma^2 u)$. Collecting coefficients proves Eq. (34.2). Define $P'(\sigma) = 2\sigma + p$, the derivative of the polynomial. The required choices follow by inserting a constant, linear or quadratic u:

condition	$u(t)$	$y_{\mathrm{p}}(t)$
$P(\sigma) \neq 0$	$F/P(\sigma)$	$\dfrac{F}{P(\sigma)}e^{\sigma t}$
$P(\sigma) = 0,\ P'(\sigma) \neq 0$	$Ft/P'(\sigma)$	$\dfrac{F}{P'(\sigma)}te^{\sigma t}$
$P(\sigma) = P'(\sigma) = 0$	$Ft^2/2$	$\dfrac{F}{2}t^2e^{\sigma t}$.

In the first row only the $P(\sigma)u$ term remains, equal to F. In the second row only $P'(\sigma)u'$ remains, again F. In the last row $u'' = F$ supplies the forcing. The last condition says that σ is a double root of the quadratic. These cases cover every complex σ, including $F = 0$. A root value of σ requires the additional power of t because $e^{\sigma t}$ alone then solves the homogeneous equation and cannot produce a nonzero forcing.

Example 34.4 (Forcing at a simple characteristic root)**.** Solve

$$y'' - 3y' + 2y = 2e^t, \qquad y(0) = 0, \quad y'(0) = 0.$$

Here $P(1) = 0$ and $P'(1) = -1$, so the particular solution is $y_{\mathrm{p}} = -2te^t$. The complete solution is $y = -2te^t + c_1e^t + c_2e^{2t}$. Its initial data give

$$c_1 + c_2 = 0, \qquad -2 + c_1 + 2c_2 = 0, \qquad c_1 = -2, \quad c_2 = 2.$$

Therefore the unique solution is $y(t) = -2(1 + t)e^t + 2e^{2t}$ on $\mathbb{R}$. The particular solution alone has the wrong initial derivative; adding the homogeneous part corrects both initial data.

34.5 The exponential series for a complex argument

Fix $z \in \mathbb{C}$ and let time t be real. Define

$$H(t) = \sum_{n=0}^{\infty} \frac{z^n t^n}{n!},$$

with constant term 1. For $z \neq 0$ and $t \neq 0$, the absolute terms have ratio $|zt|/(n+1) \to 0$. If either is zero, only the constant term remains. Thus the series converges absolutely at every real time. The real and imaginary coefficient series have absolute terms bounded by $|z|^n|t|^n/n!$, so both real power series have infinite radius. Lecture 31's differentiation theorem applies separately to their sums. Combining the component derivatives gives

$$H'(t) = \sum_{n=1}^{\infty} \frac{z^n t^{n-1}}{(n-1)!} = z \sum_{m=0}^{\infty} \frac{z^m t^m}{m!} = zH(t), \qquad H(0) = 1.$$

Reindexing and multiplication by the fixed constant z preserve these convergent sums, as follows by doing both operations at finite cutoffs and passing to the component limits. The complex first-order uniqueness theorem in Lecture 33 now gives $H(t) = e^{zt}$ for every $t \in \mathbb{R}$. At $t = 1$ this proves the general factorial-series identity

$$e^z = \sum_{n=0}^{\infty} \frac{z^n}{n!} \qquad (z \in \mathbb{C}). \tag{34.3}$$

This agrees with Lecture 16's definition $e^{a+ib} = e^a(\cos b + i \sin b)$. The imaginary-axis proof in Lecture 31 is its special case. The argument used real-parameter derivatives and first-order uniqueness, not an unproved exchange of products of infinite series.

Summary 34.

- Theorem 34.1 proves every solution of $y'' + py' + qy = 0$, for real constants p, q, belongs to the distinct-root or repeated-root family. Two initial data fix its coefficients uniquely on $\mathbb{R}$.
- The sign of $p^2 - 4q$ gives the three real forms. For a spring with $m, k > 0$ and $c \geq 0$, compare $c/(2m)$ with $\sqrt{k/m}$ to classify damping.
- For exponential forcing, Eq. (34.2) gives particular solutions with an additional factor t at a simple characteristic root and t^2 at a double root. Add the homogeneous solution to match initial data.
- Equation (34.3) holds absolutely for every complex z. Its proof differentiates real component series and applies the proved complex first-order uniqueness theorem.

Sheet 34 treats the three characteristic cases, forcing and further constant-coefficient equations. Lecture 35 applies the method to sinusoidal forcing and describes Fourier coefficients for periodic functions.

References

G. F. Simmons, *Differential Equations with Applications and Historical Notes*, 3rd ed., CRC Press, 2017, Chapters 3–4. G. Teschl, *Ordinary Differential Equations and Dynamical Systems*, Graduate Studies in Mathematics 140, AMS, 2012, §3.1 (matrix exponential, including the scalar case).